A unit-step walk in fourteen dimensions

Shallit [2] constructed an infinite standard-basis walk in $\mathbb{N}^{16}$ with no three collinear vertices. A small change to the offsets in the construction of Cambie and Kalviainen [1] lowers the dimension to fourteen. Here $\mathbb{N} = \{0, 1, 2, \ldots\}$.

Theorem. *There is an infinite sequence $P_0, P_1, \ldots \in \mathbb{N}^{14}$, starting at the origin, whose steps are standard basis vectors and no three of whose vertices are collinear.*

Proof. Let $s_2(n)$ count the 1's in the binary expansion of n, and set $t_n = s_2(n) \bmod 4 \in \{0, 1, 2, 3\}$ and $Z_n = \sum_{0 \le j < n} \mathrm{i}^{t_j}$. Choose the four offsets

$$(c_0, c_1, c_2, c_3) = (0, -1, -1 + \mathrm{i}, -\mathrm{i}), \qquad W_n = 2Z_n + c_{t_n}, \qquad H_n = 4n + t_n,$$

and identify $Q_n = (W_n, H_n)$ with $(\Re W_n, \Im W_n, H_n) \in \mathbb{Z}^3$. Writing ν_2 for the exponent of 2, we claim that for every $m < n$,

$$\nu_2\big(|W_n - W_m|^2\big) = \nu_2(H_n - H_m). \tag{1}$$

For equal states, [1, eq. (2)] gives $\nu_2(|Z_n - Z_m|^2) = \nu_2(n - m)$, so both sides are $2 + \nu_2(n - m)$. For states of opposite parity, $W_n - W_m$ has exactly one odd coordinate and $H_n - H_m$ is odd, so both sides are zero. For distinct states of equal parity, both coordinates of $W_n - W_m$ are odd and $H_n - H_m \equiv 2 \pmod 4$, so both sides are one. In particular, $W_n \neq W_m$.

The heights strictly increase, since $H_{n+1} - H_n \ge 1$. If Q_a, Q_b, Q_c were collinear, with $a < b < c$, put $A = H_b - H_a > 0$ and $B = H_c - H_b > 0$. The three complex slopes

$$\frac{W_b - W_a}{A} = \frac{W_c - W_b}{B} = \frac{W_c - W_a}{A + B}$$

would coincide. Extend ν_2 to positive rationals by $\nu_2(a/b) = \nu_2(a) - \nu_2(b)$. By (1), their squared moduli have valuations $-\nu_2(A)$, $-\nu_2(B)$ and $-\nu_2(A + B)$, respectively. Thus $A, B, A + B$ would have the same valuation, which is impossible: after removing their common power of 2, the first two are odd but their sum is even.

The step corresponding to $(t_n, t_{n+1}) = (r, s)$ is $v_{r,s} = (2\mathrm{i}^r + c_s - c_r,\ 4 + s - r)$. There are at most fourteen such vectors, since

$$v_{0,2} = v_{1,3} = (1 + \mathrm{i}, 6), \qquad v_{2,0} = v_{3,1} = (-1 - \mathrm{i}, 2).$$

List the distinct vectors as $v_1, \ldots, v_k$, where $k \le 14$. Starting at $P_0 = 0$, replace each step v_j of (Q_n) by the jth standard basis vector of $\mathbb{R}^{14}$. This defines a unit-step walk in $\mathbb{N}^{14}$, with any unused coordinates zero. The linear map $T : \mathbb{R}^{14} \to \mathbb{R}^3$, $T(x) = \sum_{j=1}^{k} x_j v_j$, satisfies $T(P_n) = Q_n$: both sides are the sum of the first n steps of (Q_n), since $Q_0 = 0$.

Finally, suppose P_a, P_b, P_c were collinear, where $a < b < c$. The coordinate sum of P_n is n, so these points are distinct. Write $P_b = (1 - \lambda)P_a + \lambda P_c$. Taking coordinate sums gives $b = (1 - \lambda)a + \lambda c$, hence $\lambda = (b - a)/(c - a) \in (0, 1)$. Applying T now yields

$$Q_b = T(P_b) = (1 - \lambda)T(P_a) + \lambda T(P_c) = (1 - \lambda)Q_a + \lambda Q_c.$$

Since $H_a < H_b < H_c$, the three image points are distinct, contradicting the absence of collinear triples among the Q_n. $\square$

References

[1] S. Cambie and E. Kalviainen, *An infinite small-step $\mathbb{Z}^3$-walk with no collinear triple*, arXiv:2609.01766v1, 2026.

[2] J. Shallit, *An infinite walk in $\mathbb{N}^{16}$, using only unit steps, with no three collinear points*, manuscript, September 4, 2026.